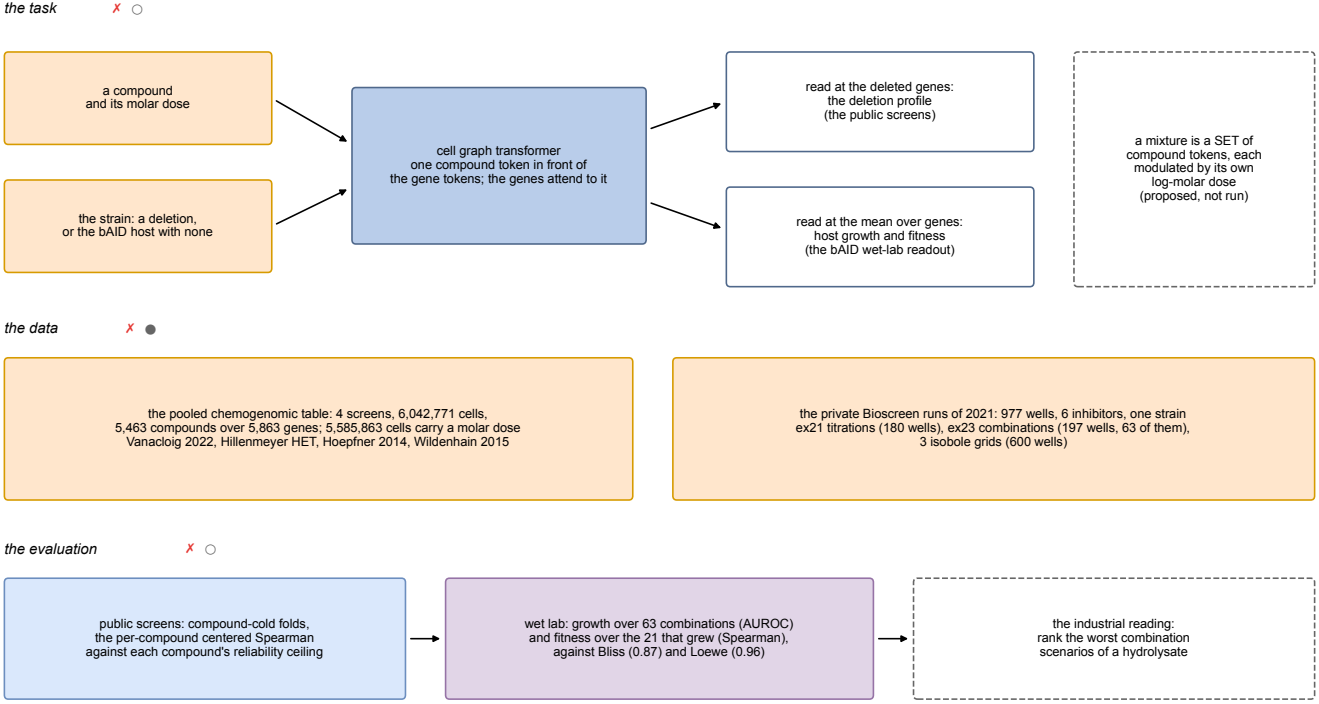
the task ✗ ○
a compound
and its molar dose
the strain: a deletion,
or the bAID host with none
cell graph transformer
one compound token in front of
the gene tokens; the genes attend to it
read at the deleted genes:
the deletion profile
(the public screens)
read at the mean over genes:
host growth and fitness
(the bAID wet-lab readout)
a mixture is a SET of
compound tokens, each
modulated by its own
log-molar dose
(proposed, not run)
the data ✗ ●
the pooled chemogenomic table: 4 screens, 6,042,771 cells,
5,463 compounds over 5,863 genes; 5,585,863 cells carry a molar dose
Vanacloig 2022, Hillenmeyer HET, Hoepfner 2014, Wildenhain 2015
the private Bioscreen runs of 2021: 977 wells, 6 inhibitors, one strain
ex21 titrations (180 wells), ex23 combinations (197 wells, 63 of them),
3 isobole grids (600 wells)
the evaluation ✗ ○
public screens: compound-cold folds,
the per-compound centered Spearman
against each compound's reliability ceiling
wet lab: growth over 63 combinations (AUROC)
and fitness over the 21 that grew (Spearman),
against Bliss (0.87) and Loewe (0.96)
the industrial reading:
rank the worst combination
scenarios of a hydrolysate
section status ✗ not done, agents may edit freely
author-reviewed
✓ final
figure status ● on file
◐ partial
○ idea only